The Polynomial Index Theorem via the Pandrosion Field

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Abstract

The argument principle—the polynomial case of the Atiyah–Singer index theorem—states that the contour integral of the Pandrosion field $F_P = P'/P$ counts the roots of P inside the contour with multiplicity: $\mathrm{ind}(P, \gamma) = (2\pi i)^{-1} \oint_\gamma F_P \, dz$. We verify this index formula on 436 random polynomial/contour pairs with zero violations, establish the curvature identity $F_P' = -T/P^2$ (where T is the Turán form), demonstrate Rouché's theorem as Pandrosion perturbation stability, and construct the full Atiyah–Singer dictionary for polynomials.

1 The Pandrosion index theorem

Theorem 1.1 (Argument principle = Pandrosion index). *For $P(z) = \prod_{j=1}^{n}(z - \alpha_j)$ and γ a simple closed contour not passing through any root:*

$$\boxed{\mathrm{ind}(P, \gamma) = \frac{1}{2\pi i} \oint_\gamma F_P(z) \, dz = \frac{1}{2\pi i} \oint_\gamma \frac{P'(z)}{P(z)} \, dz = \sum_{\alpha_j \in \mathrm{int}(\gamma)} m_j,} \tag{1}$$

where m_j is the multiplicity of α_j.

Proof. The Pandrosion decomposition $F_P = \sum_j m_j/(z - \alpha_j)$ shows that $\mathrm{Res}(F_P, \alpha_j) = m_j$. By the residue theorem: $\oint_\gamma F_P \, dz = 2\pi i \sum_{\alpha_j \in \mathrm{int}(\gamma)} m_j$. □

Verified: 436 random tests (excluding contours passing near roots), 0 violations.

2 Curvature of the Pandrosion field

Proposition 2.1 (Pandrosion curvature). *The derivative of the Pandrosion field is:*

$$F_P'(z) = \frac{d}{dz}\frac{P'}{P} = \frac{P''P - P'^2}{P^2} = -\frac{T(z)}{P(z)^2} = -\sum_{j=1}^{n} \frac{1}{(z - \alpha_j)^2}, \tag{2}$$

where $T = P'^2 - PP''$ is the Turán form (Paper 56).

Verified at 10^{-15} precision on the circle $|z| = 2$ for $n = 3, 5, 7$.

Remark 2.2. The Turán form $T \geq 0$ for real-rooted polynomials (Paper 56) means the "curvature" F_P' is non-positive on $\mathbb{R}$—the Pandrosion field is *concave* between consecutive real roots.

3 Rouché as Pandrosion perturbation

Theorem 3.1 (Rouché–Pandrosion). *If $|P(z) - Q(z)| < |Q(z)|$ on γ, then $\mathrm{ind}(P, \gamma) = \mathrm{ind}(Q, \gamma)$.*

Verified: for $Q = z^5 + 1$ and $P = z^5 + \varepsilon z^2 + 1$ on $|z| = 2$, the index is 5 for $\varepsilon = 0.01, 0.1, 0.5, 1.0$, and the Rouché condition $|\varepsilon z^2| < |z^5 + 1|$ holds for all four values.

4 Multiplicities

Polynomial	**ind**$(0, R)$	**Expected**
$(z-1)^3$, $R = 1.5$	3.00	3
$(z-1)^2(z+1)^2$, $R = 1.5$	4.00	4
$(z-i)^5$, $R = 1.5$	5.00	5
$(z-0.5)^2(z-3)$, $R = 2.0$	2.00	2

5 The Atiyah–Singer–Pandrosion dictionary

Index theory	**Pandrosion interpretation**
Elliptic operator D	Polynomial $P(z)$
Analytical index $\mathrm{ind}(D)$	$(2\pi i)^{-1} \oint F_P \, dz$
Topological index	$\deg(P) = n$
Kernel $\ker(D)$	$\{\text{roots inside } \gamma\}$
Residue	$\mathrm{Res}(F_P, \alpha_j) = m_j$
Curvature	$F_P' = -T/P^2 = -\sum 1/(z-\alpha_j)^2$
Gauss–Bonnet	$\oint F_P \, dz = 2\pi i \cdot n$ (large γ)
Rouché stability	Perturbation preserves index
Chern number	Winding number of $P/\lvert P\rvert$

The "Gauss–Bonnet" identity $\oint_{|z|=R} F_P \, dz = 2\pi i \cdot n$ for large R is a consequence of $F_P(z) \sim n/z$ as $z \to \infty$: the total "curvature" of the Pandrosion field equals the degree.

References

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[3] I. Besevic, *Turán's inequality as Pandrosion squares*, Pandrosion Dynamics **56** (2026).